

HW 5 #3 Use Master's Method for Bonus solve explicitly
 - $3n^2 + 2n + 3$ - Master's method would only give Θ class - $\Theta(n^2)$

#10 $x^y = x + x^{y-1}$

$x^n = x + x^{n-1}$

$T(n) = T(n-1) + 1$

#11

$$x^n = \begin{cases} x^{n/2} + x^{n/2}, & x \text{ is even} \\ x^{\lfloor n/2 \rfloor} + x^{\lfloor n/2 \rfloor}, & x \text{ is odd} \end{cases}$$

$$T(n) = \begin{cases} T(n/2) + T(n/2) + 2 \\ 2T(n/2) + 2 \end{cases}$$